METHANOL CONVERTER VER 1

Operator Training Simulator · R-201 quench converter and synthesis loop

Chulalongkorn University · Department of Chemical Engineering

PLOT PLAN, ELEVATIONS AND PIPING ISOMETRICS

Synthesis loop R-201 · drawing rev A · teaching layout

DOCUMENT NO. : MEOH-V1-LAY-001

DOCUMENT TITLE : Plot Plan, Elevations and Piping Isometrics

PROJECT : Methanol Converter Ver 1 · OTS

STATUS : Rev 0 · Issued for teaching

Teaching document generated from the Methanol Converter Ver 1 simulator model; numbers come from the model and the OTS tabs. Not for construction.

Rev.	Date	Revision status	Originator	Checker	Approver
0	27-Oct-26	Issued for teaching	A. Soottitantawat		

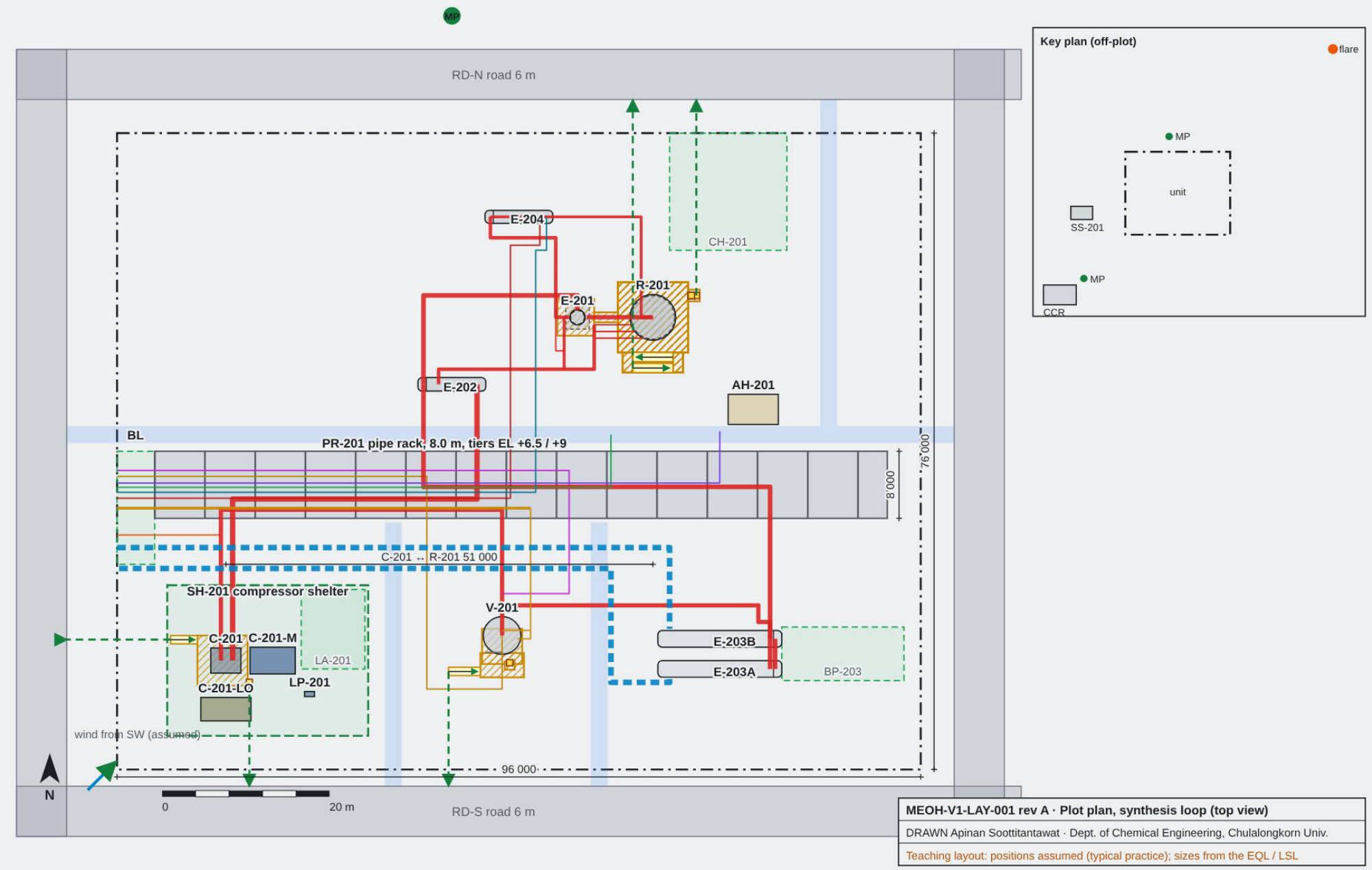
1 Purpose and basis

This document is the layout deliverable of the R-201 synthesis loop: the plot plan, area plans with every local gauge and hand valve, two elevations and a V-201 detail, the equipment-location table, six piping isometrics with bills of materials, and the layout checks. There are no real layout drawings for this unit: equipment sizes come from the equipment list MEOH-V1-ME-EQL-001 and its datasheets, line numbers and sizes from MEOH-PID-001/002 and the line sizing list MEOH-V1-PE-LSL-001, instruments from MEOH-V1-IN-INL-001, hazardous zones from MEOH-V1-EE-HAC-001. Every position, elevation, route and spacing rule is assumed (typical practice).

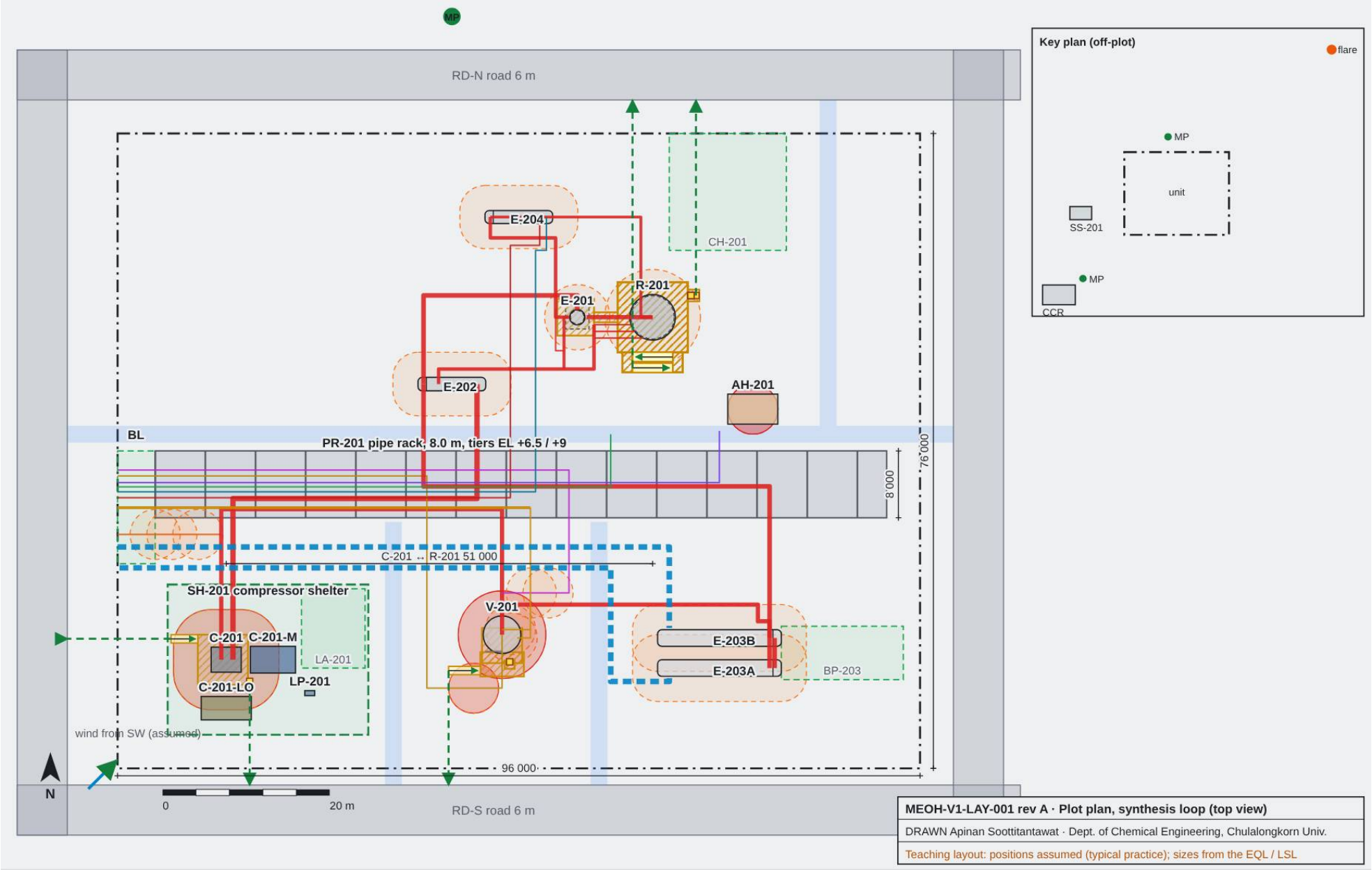
Item	Value
Drawing	MEOH-V1-LAY-001 rev A — Plot plan, elevations and piping isometrics — synthesis loop R-201
Coordinates	south-west corner of the unit plot (x 0, y 0); x east, y north, z up from EL 0.000 = top of paving; metres
Prevailing wind	from the south-west (225°) — assumed prevailing wind (SW monsoon, tropical site); confirm with the site wind rose
Data	layout.json: 16 equipment items, 28 lines, 63 instruments, 23 operables, 19 platforms and landings
Basis note	Teaching layout: equipment sizes from MEOH-V1-ME-EQL-001 and the datasheets, line numbers and sizes from MEOH-V1-PE-LSL-001 / MEOH-PID-001/002; every position, elevation and route is assumed (typical practice).

The same data drive the Layout & isometrics tab and the Visual plant 3D tab of the OTS; the checks in section 6 are computed from them.

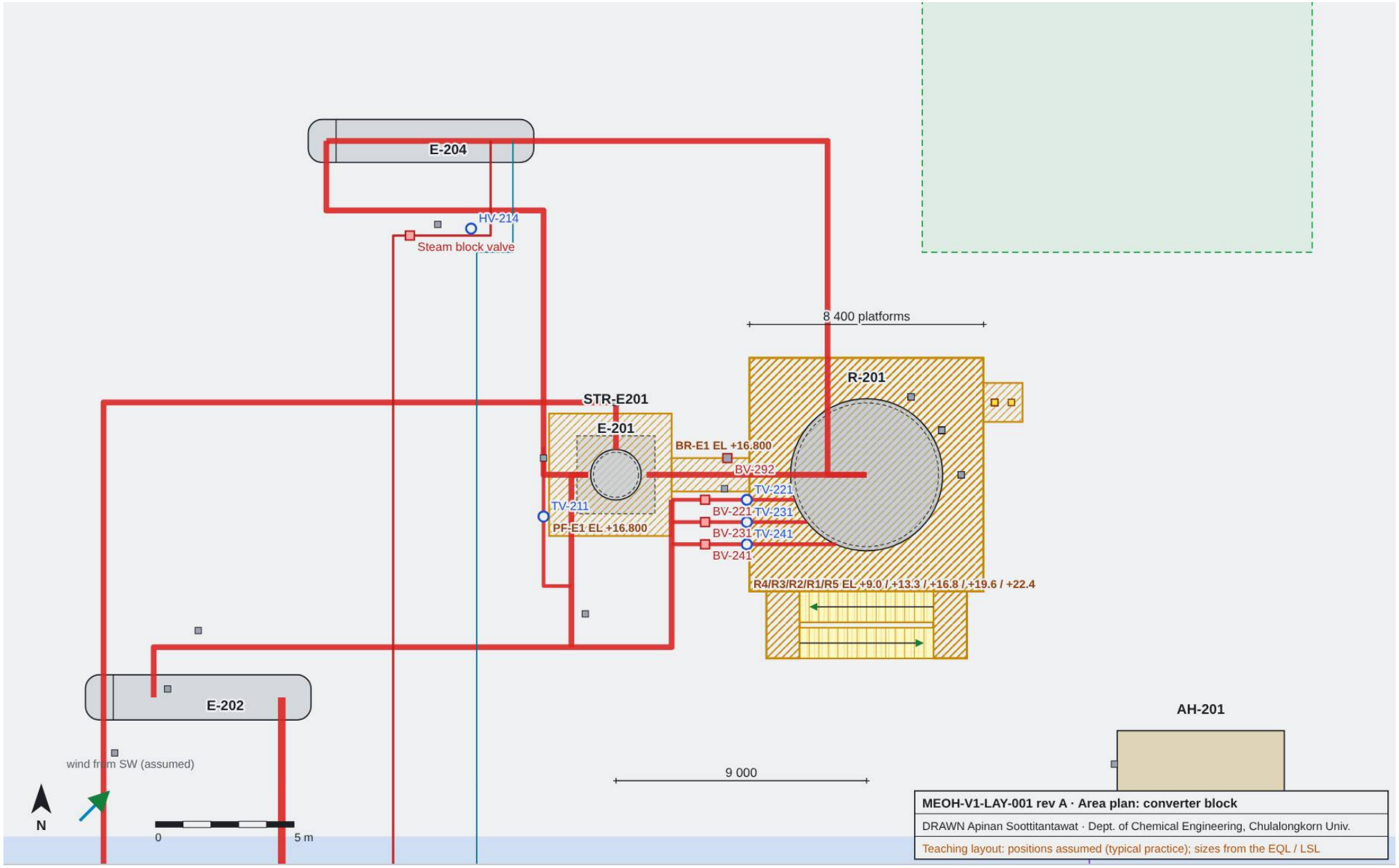
2 Plot plan


MP
RD-N road 6 m
E-204
CH-201
R-201
E-201
E-202
AH-201
BL
PR-201 pipe rack: 8.0 m, tiers EL +6.5 / +9
8 000
76 000
C-201 ↔ R-201 51 000
SH-201 compressor shelter
V-201
C-201
C-201-M
LA-201
C-201-LO
LP-201
E-203B
E-203A
BP-203
wind from SW (assumed)
96 000
N
0
20 m
RD-S road 6 m
Key plan (off-plot)
flare
MP
unit
SS-201
MP
CCR
MEOH-V1-LAY-001 rev A · Plot plan, synthesis loop (top view)
DRAWN Apinan Soottitantawat · Dept. of Chemical Engineering, Chulalongkorn Univ.
Teaching layout: positions assumed (typical practice); sizes from the EQL / LSL

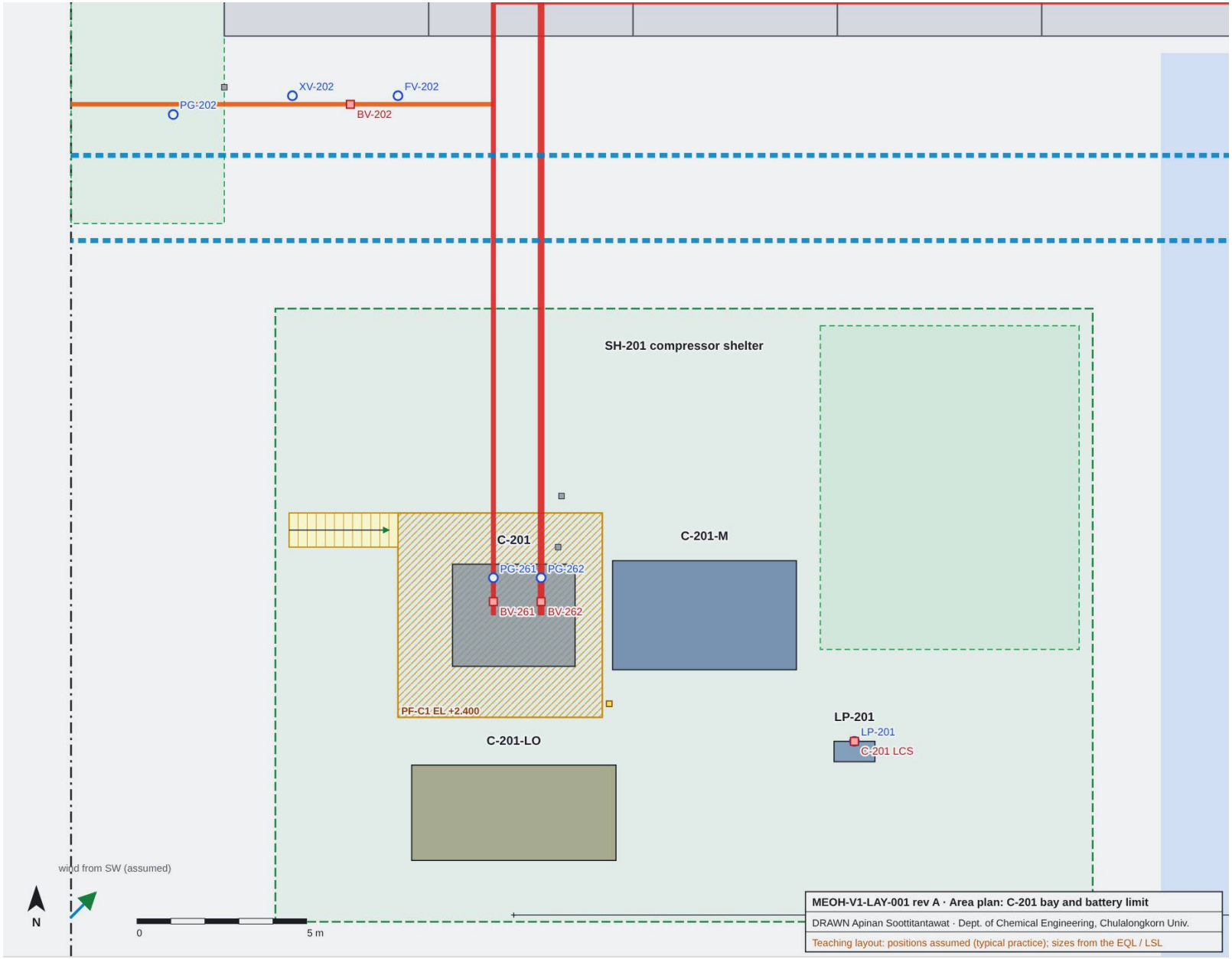
Plot plan with the hazardous areas of MEOH-V1-EE-HAC-001 (Zone 2 R 3 m, 4.5 m round C-201; Zone 1 at the analyser vents)



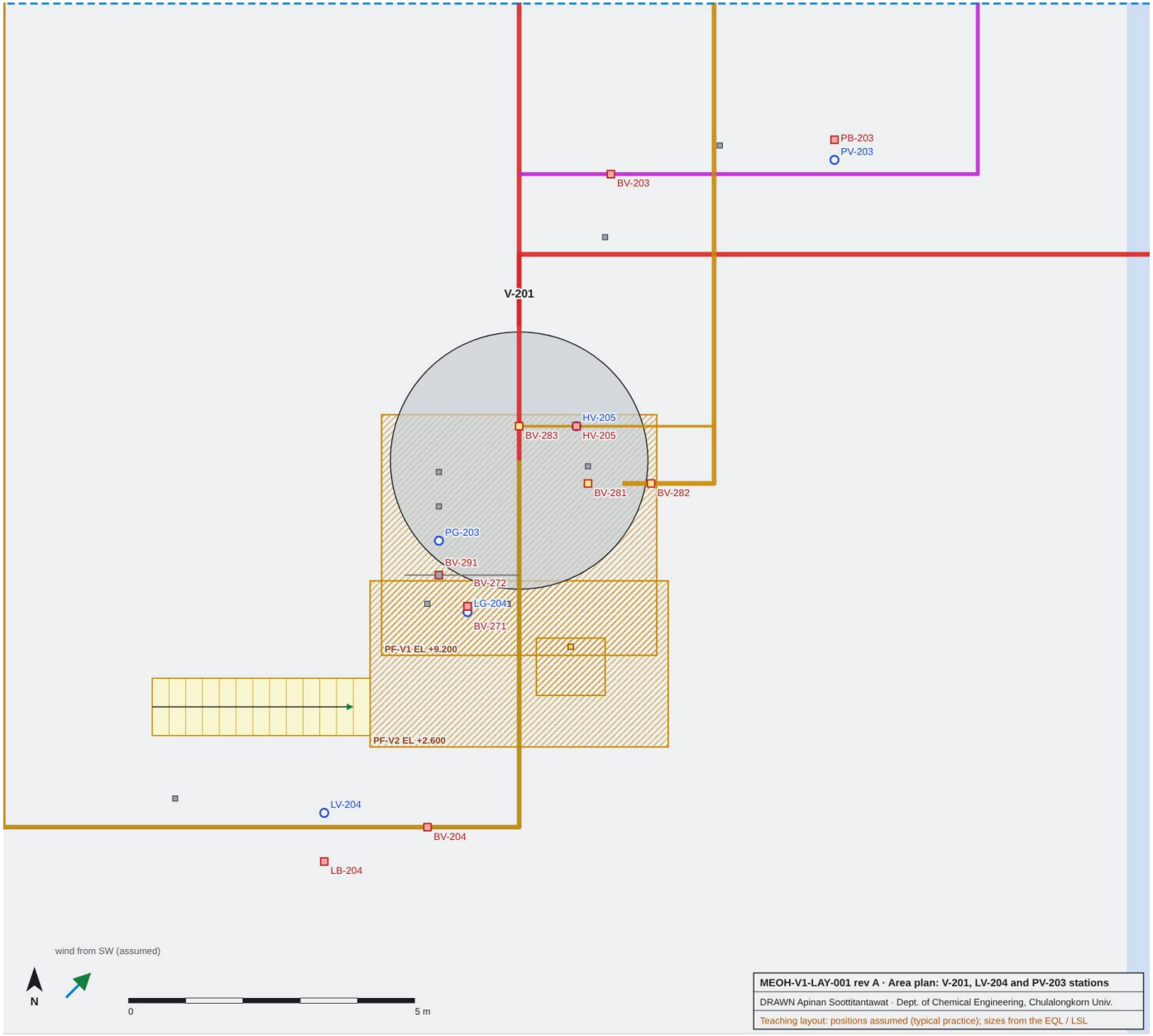
Area plan: converter block — blue circles local gauges, red squares hand valves, grey squares transmitters



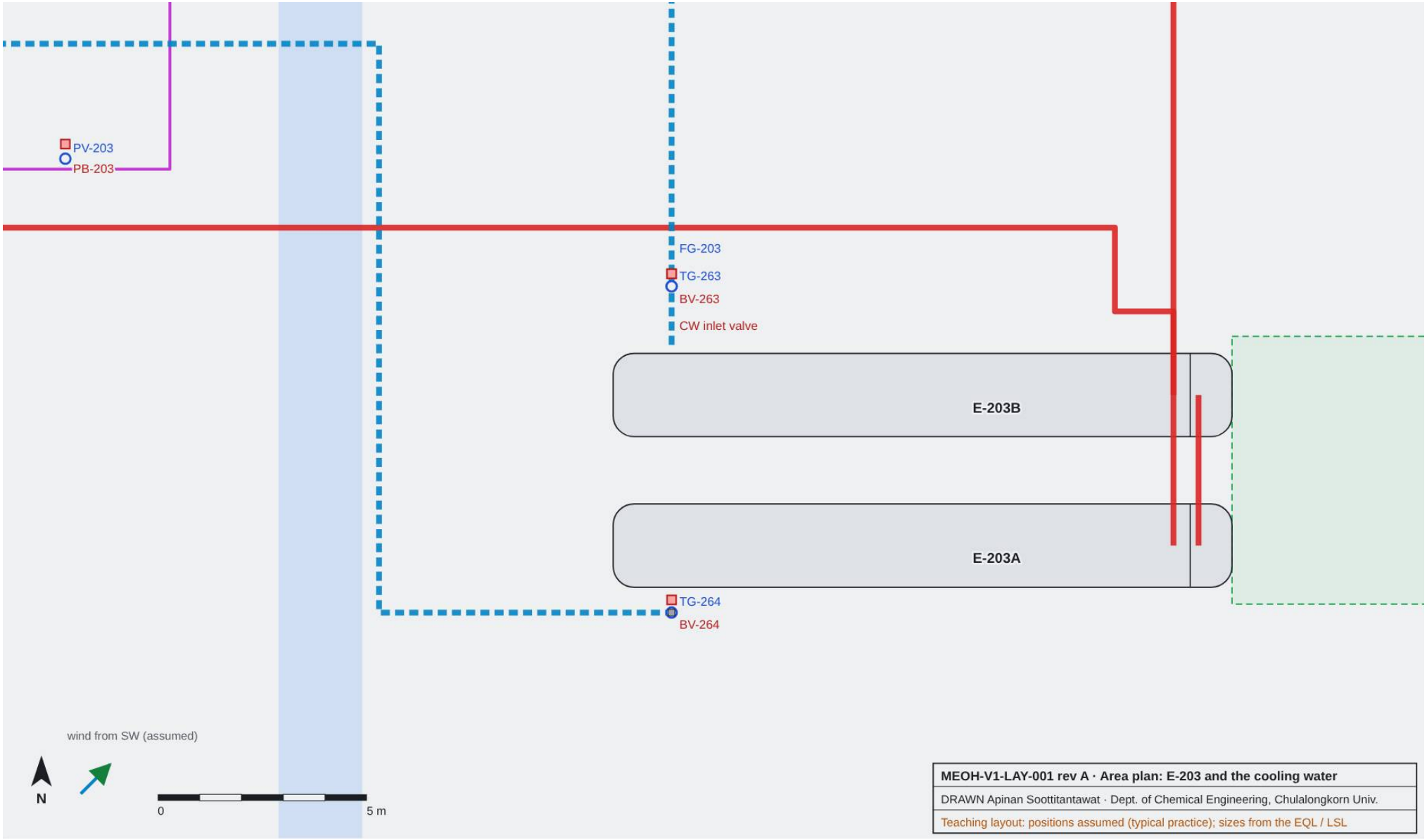
Area plan: C-201 bay and battery limit — blue circles local gauges, red squares hand valves, grey squares transmitters



Area plan: V-201, LV-204 and PV-203 stations — blue circles local gauges, red squares hand valves, grey squares transmitters



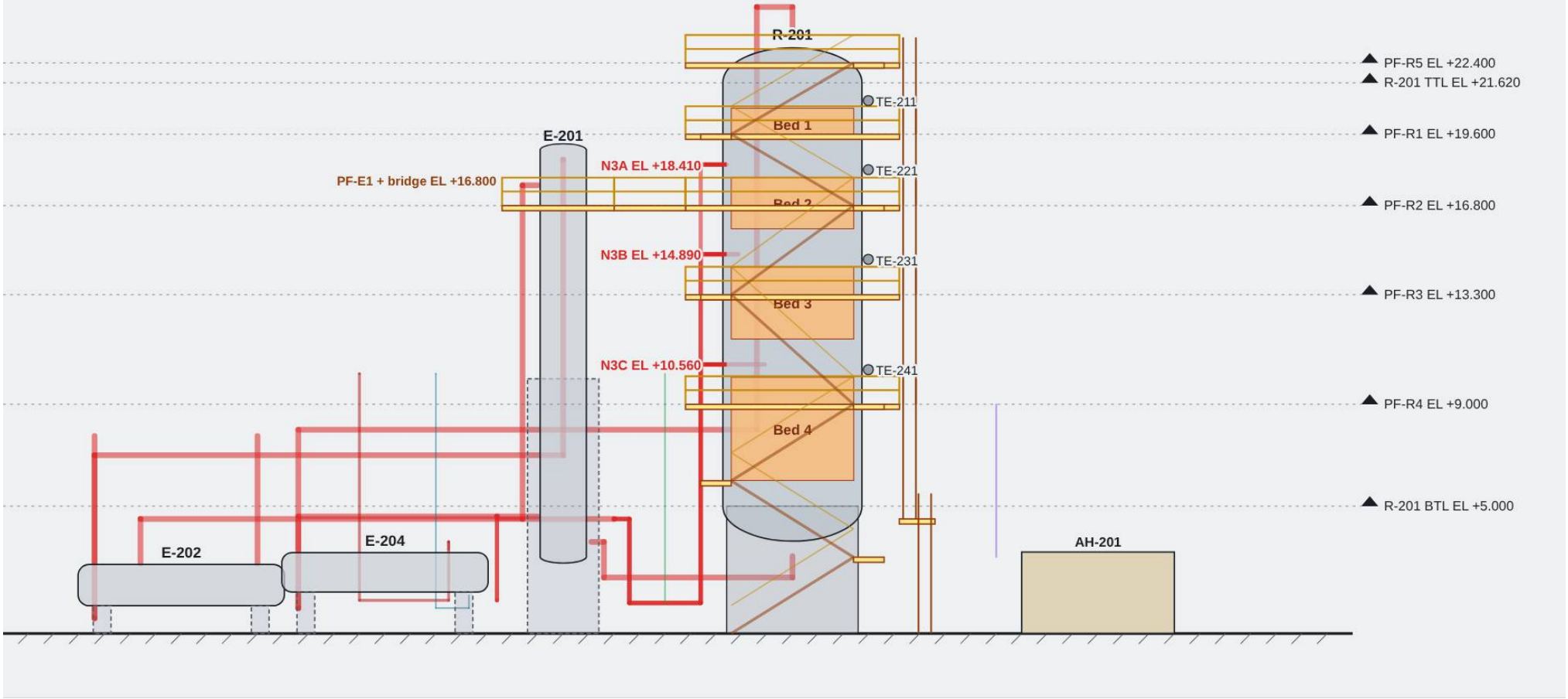
Area plan: E-203 and cooling water — blue circles local gauges, red squares hand valves, grey squares transmitters



3 Elevations and levels

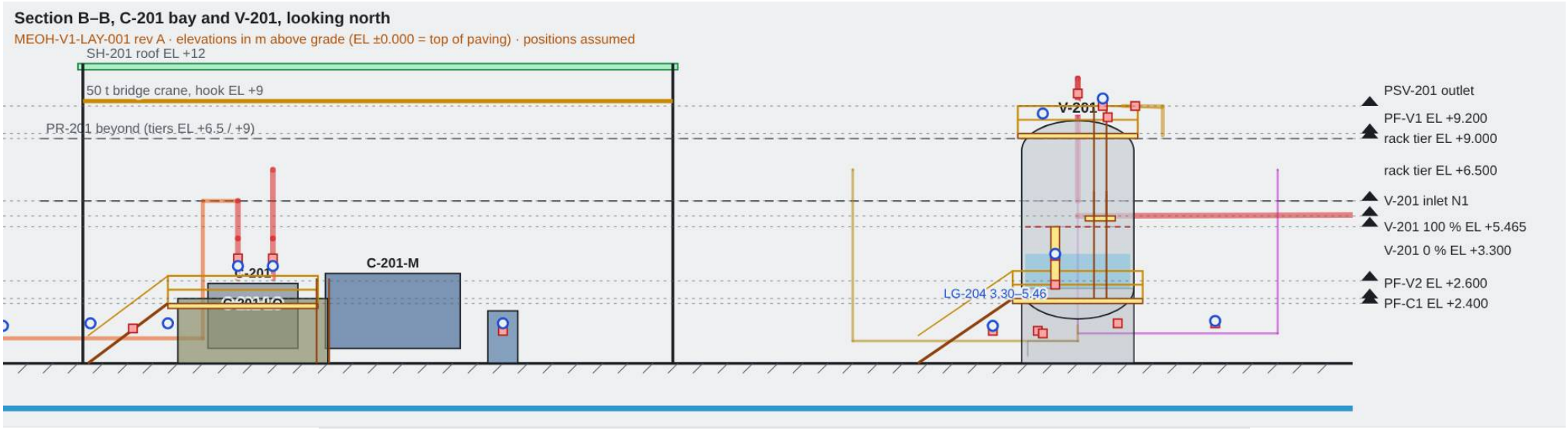
Section A–A, converter block, looking north (east → right)

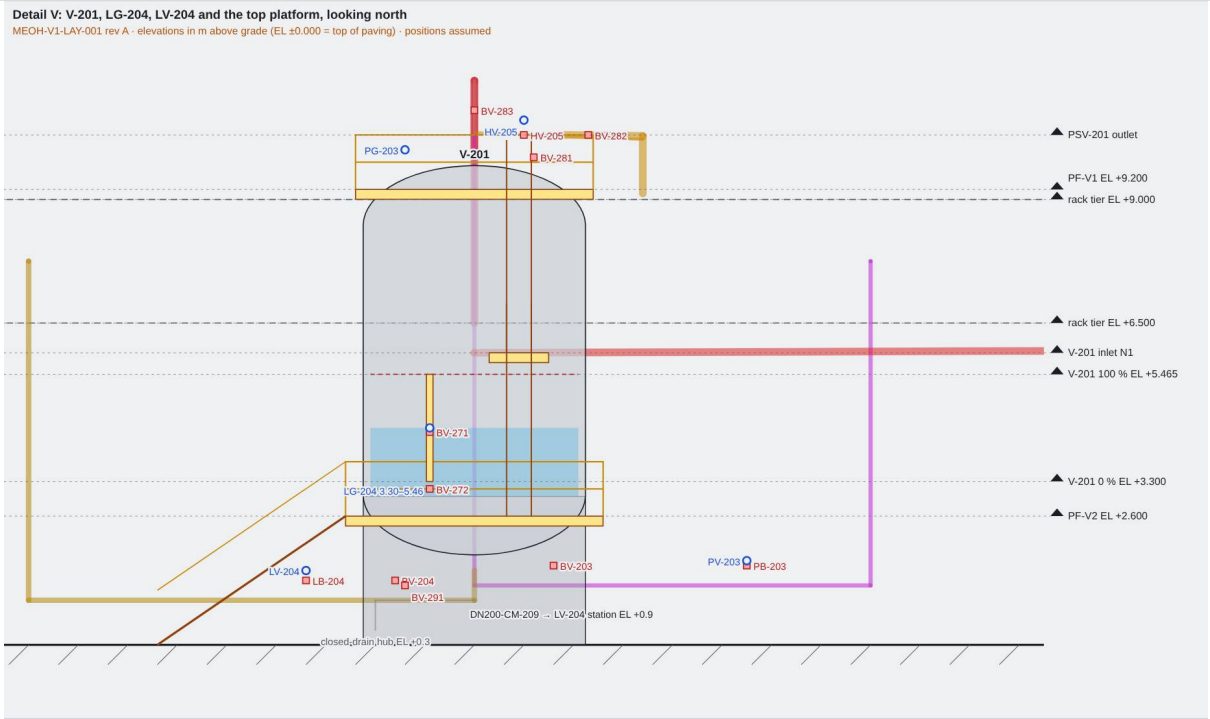
MEOH-V1-LAY-001 rev A · elevations in m above grade (EL ±0.000 = top of paving) · positions assumed



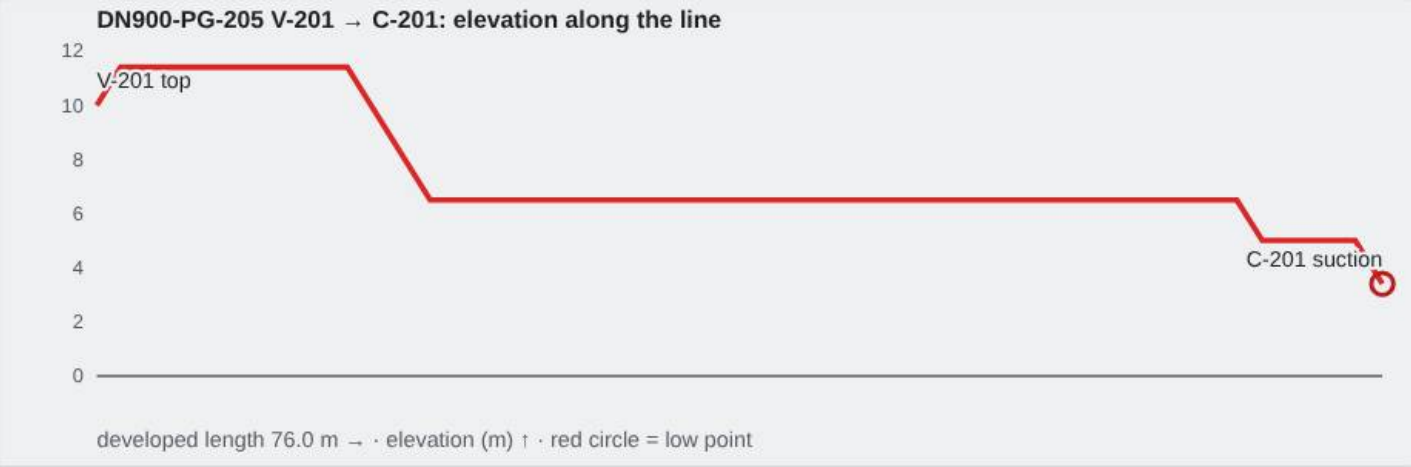
R-201 platform levels against the bed nozzles (DS-001 stack: 1.0 m top space, beds 1.21 · 2.02 · 2.83 · 4.05 m, quench zones 1.5 m)

Platform	Floor	Nozzle / instrument	Elevation	Above the floor (m)
PF-R4 Bed 4 platform	EL +9.000	N3C quench (TV-241)	EL +10.560	1.56
		TE-241 bed inlet	EL +10.360	1.36
		T4 multipoint TE-242…244	EL +9.760	0.76
PF-R3 Bed 3 platform	EL +13.300	N3B quench (TV-231)	EL +14.890	1.59
		TE-231 bed inlet	EL +14.690	1.39
		T3 multipoint TE-232…234	EL +14.090	0.79
PF-R2 Bed 2 platform	EL +16.800	N3A quench (TV-221)	EL +18.410	1.61
		TE-221 bed inlet	EL +18.210	1.41
		T2 multipoint TE-222…224	EL +17.610	0.81
PF-R1 Bed 1 platform	EL +19.600	TE-211 bed inlet	EL +20.920	1.32
		T1 multipoint TE-212…214	EL +20.320	0.72
PF-R5 Top platform (manways, N1)	EL +22.400	Head apex, manways M1/M2	EL +22.992	0.59


Section B–B, C-201 bay and V-201, looking north
MEOH-V1-LAY-001 rev A · elevations in m above grade (EL ±0.000 = top of paving) · positions assumed
SH-201 roof EL +12
50 t bridge crane, hook EL +9
PR-201 beyond (tiers EL +6.5 / +9)
C-201-M
V-201
LG-204 3.30–5.46
PSV-201 outlet
PF-V1 EL +9.200
rack tier EL +9.000
rack tier EL +6.500
V-201 inlet N1
V-201 100 % EL +5.465
V-201 0 % EL +3.300
PF-V2 EL +2.600
PF-C1 EL +2.400


Detail V: V-201, LG-204, LV-204 and the top platform, looking north
MEOH-V1-LAY-001 rev A · elevations in m above grade (EL ±0.000 = top of paving) · positions assumed
BV-283
HV-205
HV-205
BV-282
PG-203
V-201
BV-281
BV-271
LG-204 3.30–5.46
BV-272
LV-204
LB-204
PV-204
BV-291
BV-203
PV-203
PB-203
DN200-CM-209 → LV-204 station EL +0.9
closed drain hub, EL +0.3
PSV-201 outlet
PF-V1 EL +9.200
rack tier EL +9.000
rack tier EL +6.500
V-201 inlet N1
V-201 100 % EL +5.465
V-201 0 % EL +3.300
PF-V2 EL +2.600

DN900-PG-205: elevation along the line (low point at the C-201 suction nozzle, EQ-ly-02)



4 Equipment-location table

Tag	Item	x (m)	y (m)	z bottom	z top	Size	Source
R-201	Methanol quench converter	64.00	54.00	3.63	22.99	OD 5160 (+150 ins.) × T/T 16.62 m	EQL/DS-001: ID 4810, T/T 16 620, shell 175 mm; 150 mm hot insulation and 5.0 m skirt assumed
E-201	Feed/effluent interchanger (hot), vertical	55.00	54.00	3.00	19.00	OD 1616 (+100 ins.) × T/T 16 m	EQL/DS-003: shell ID 1500, tubes 12.2 m → 16.0 m overall with channels (assumed); on a steel structure, bottom EL +3.0
E-202	Feed/effluent interchanger (cold)	40.00	46.00	1.09	2.71	OD 1462 × 8.1 m, axis x, channel W	EQL/DS-004: shell ID 1350, U-tubes 6.1 m → 8.1 m overall (assumed); saddles at grade
E-203A	Final cooler (CW), shell A (process in)	72.00	12.00	1.21	3.20	OD 1990 × 14.8 m, axis x, channel E	EQL/DS-005: 2 shells in series, ID 1850, U-tubes 12.2 m → 14.8 m overall (assumed)
E-203B	Final cooler (CW), shell B (process out)	72.00	15.60	1.21	3.20	OD 1990 × 14.8 m, axis x, channel E	EQL/DS-005: second shell, side by side with A, 1.6 m clear (assumed)
E-204	Start-up heater (HP steam)	48.00	66.00	1.63	3.17	OD 1346 × 8.1 m, axis x, channel W	EQL/DS-006: shell ID 1250, U-tubes 6.1 m → 8.1 m overall (assumed)
V-201	HP separator	46.00	16.00	1.81	9.70	OD 4490 × T/T 5.5 m	EQL/DS-002: ID 4200, T/T 5500, shell 145 mm; bottom tangent EL +3.0 = RLB fire case (MEOH-V1-PE-RLB-001)
C-201	Loop circulator (barrel casing)	13.00	13.00	0.60	3.20	3.6 × 3 × 2.6 m	EQL/DS-007 (1928 kNm³/h, 6.53 MW at 80 %); casing size assumed; top suction and discharge nozzles, block foundation
C-201-M	C-201 motor 14 MW, 11 kV (VSD fed)	18.60	13.00	0.60	3.60	5.4 × 3.2 × 3 m	EE-LL L01: 14 000 kW rating; frame size assumed; Ex-certified for the C-201 Zone 2 (HAC RS-05)
C-201-LO	C-201 lube-oil and seal-gas console	13.00	7.20	0.00	2.60	6 × 2.8 × 2.6 m	EE-LL L02–L07 (vendor package, assumed size)
LP-201	C-201 local control panel (speed, power, vibration; local stop)	23.00	9.00	0.00	2.10	1.2 × 0.6 × 2.1 m	assumed (local control station; Ex p purged panel in Zone 2)

Tag	Item	x (m)	y (m)	z bottom	z top	Size	Source
SH-201	C-201 compressor shelter (open sides, roof EL +12.0) with 50 t bridge crane	18.00	13.00	0.00	12.00	24 × 18 × 12 m	assumed: open-sided shelter (good ventilation, HAC RS-04/05); crane covers casing, motor and the laydown bay
STR-E201	E-201 support structure (lugs at EL +10.0)	55.00	54.00	0.00	10.00	2.8 × 2.8 × 10 m	assumed
AH-201	Analyser house (AIT-205 GC, AIT-208)	76.00	43.00	0.00	3.20	6 × 3.6 × 3.2 m	EE-LL L09/L10, HAC RS-09: purged shelter, inside Zone 2 (assumed size)
CCR	Central control room (off-plot, shared)	-60.00	-55.00	0.00	5.00	30 × 18 × 5 m	assumed: off-plot, upwind, outside the blast distance
SS-201	Substation (11 kV switchgear, C-201 VSD, 400 V MCC, UPS)	-40.00	20.00	0.00	6.00	20 × 12 × 6 m	assumed: off-plot, upwind, non-hazardous area; EE-LL cable 150 m basis

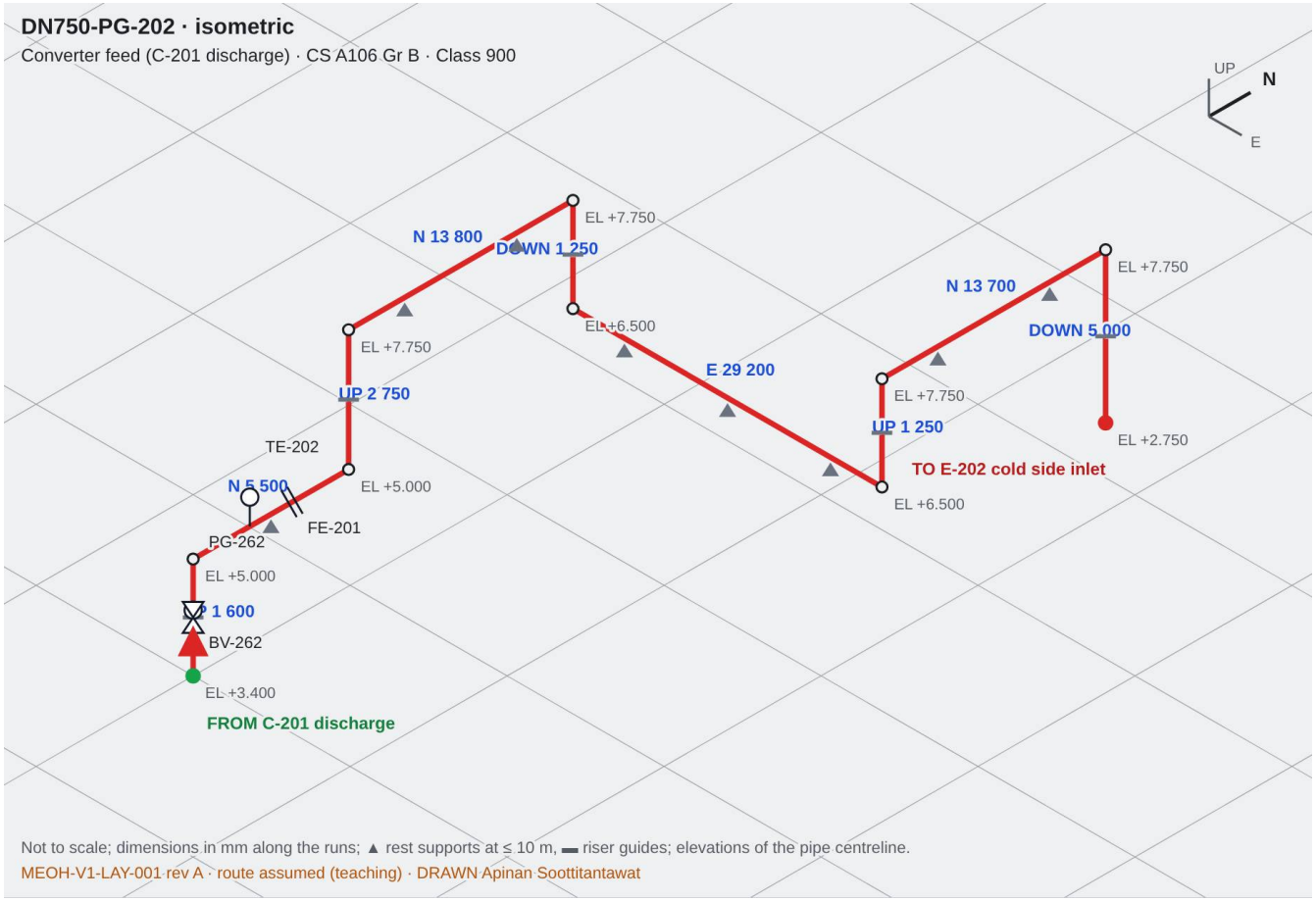
5 Piping isometrics and bills of materials

Isometrics are not drawn to scale; dimensions are true lengths in mm along the runs; elevations of the pipe centreline. Take-off rules (spans, flanges, welds) are typical and assumed.

Line	Service	From → to	Length (m)	Bends	On the P&ID
DN300-SG-202	Make-up syngas	Battery limit (syngas compressor) → C-201 suction header via FV-202, XV-202	17.9	2	yes
DN900-PG-205	V-201 vapour / recycle gas	V-201 top → C-201 suction (recycle; purge tee on the riser)	76.0	8	yes
DN750-PG-202	Converter feed (C-201 discharge)	C-201 discharge → E-202 cold side inlet	74.0	8	yes
DN150-PU-203	Purge gas	V-201 vapour riser (tee) → Reformer fuel header via PV-203, FI-203	100.2	7	yes
DN200-CM-209	Crude methanol	V-201 bottom → Let-down / distillation via LV-204, FI-209	92.0	8	yes
DN750-PG-201	Converter feed after E-202	E-202 cold side outlet (off-sheet) → Quench header / bed-1 feed branch	27.5	3	yes
DN300-PG-222	Cold shot to bed 2	Quench header → R-201 bed 2 distributor via TV-221	24.9	4	yes
DN350-PG-232	Cold shot to bed 3	Quench header → R-201 bed 3 distributor via TV-231	21.9	4	yes
DN350-PG-242	Cold shot to bed 4	Quench header → R-201 bed 4 distributor via TV-241	18.6	4	yes
DN600-PG-213	Bed 1 feed (after E-201, E-204)	E-204 outlet → R-201 top nozzle	53.7	6	yes
DN900-PG-251	Converter effluent	R-201 bottom (bed 4 outlet) → E-201 hot side inlet	10.1	3	yes
DN650-PG-X01	Bed-1 feed branch, quench header → E-201 shell inlet	DN750-PG-201 tee → E-201 shell inlet (top)	19.9	3	provisional
DN300-PG-X02	TV-211 cold-side bypass round E-201	DN650-PG-X01 → DN750-PG-X03	12.5	3	provisional
DN750-PG-X03	E-201 shell outlet → E-204 tubes	E-201 shell outlet (bottom) → E-204 channel (bottom)	25.5	5	provisional
DN1000-PG-X04	E-201 tube outlet → E-202 tubes (effluent)	E-201 top channel → E-202 channel (top)	46.6	4	provisional
DN1000-PG-X05	E-202 tube outlet → E-203A (effluent)	E-202 channel (bottom) → E-203A channel (top)	87.3	7	provisional
DN1000-PG-X06	E-203A → E-203B (two-phase)	E-203A channel (bottom) → E-203B channel (top)	7.6	3	provisional
DN900-PG-X07	E-203B → V-201 inlet (two-phase, 40 °C)	E-203B channel (bottom) → V-201 N1 (north)	42.1	5	provisional
DN300-FL-X08	PSV-201 outlet → flare sub-header	PSV-201 (V-201 top) → DN600-FL-X10	17.2	1	provisional
DN100-FL-X09	HV-205 blowdown → PSV-201 outlet line	DN900-PG-205 (V-201 vapour) → DN300-FL-X08	4.5	1	provisional
DN600-FL-X10	Flare sub-header → BL (flare KO drum off-plot)	DN300-FL-X08 → Flare header (off-plot)	49.4	0	provisional
DN1200-CW-X11	Cooling water supply → E-203 (shell side)	CW supply header (BL) → E-203B shell inlet	79.7	3	provisional
DN1200-CW-X12	Cooling water return from E-203	E-203A shell outlet → CW return header (BL)	84.0	4	provisional

Line	Service	From → to	Length (m)	Bends	On the P&ID
DN200-HS-X13	HP steam 40 bar g → E-204 shell	HP steam header (BL) → E-204 shell inlet	104.8	9	provisional
DN100-SC-X14	E-204 condensate → BL	E-204 shell outlet → Condensate header (BL)	103.1	8	provisional
DN80-NG-X15	Nitrogen (purging, start-up) → utility station UT-201	N_2 header (BL) → UT-201 near R-201	83.5	5	provisional
DN50-IA-X16	Instrument air header → AH-201 and the unit	IA header (BL) → AH-201 / valve actuators	92.2	4	provisional
DN50-CD-X17	V-201 bottom drain → closed drain	DN200-CM-209 (V-201 bottom) → Closed drain hub	2.6	1	provisional

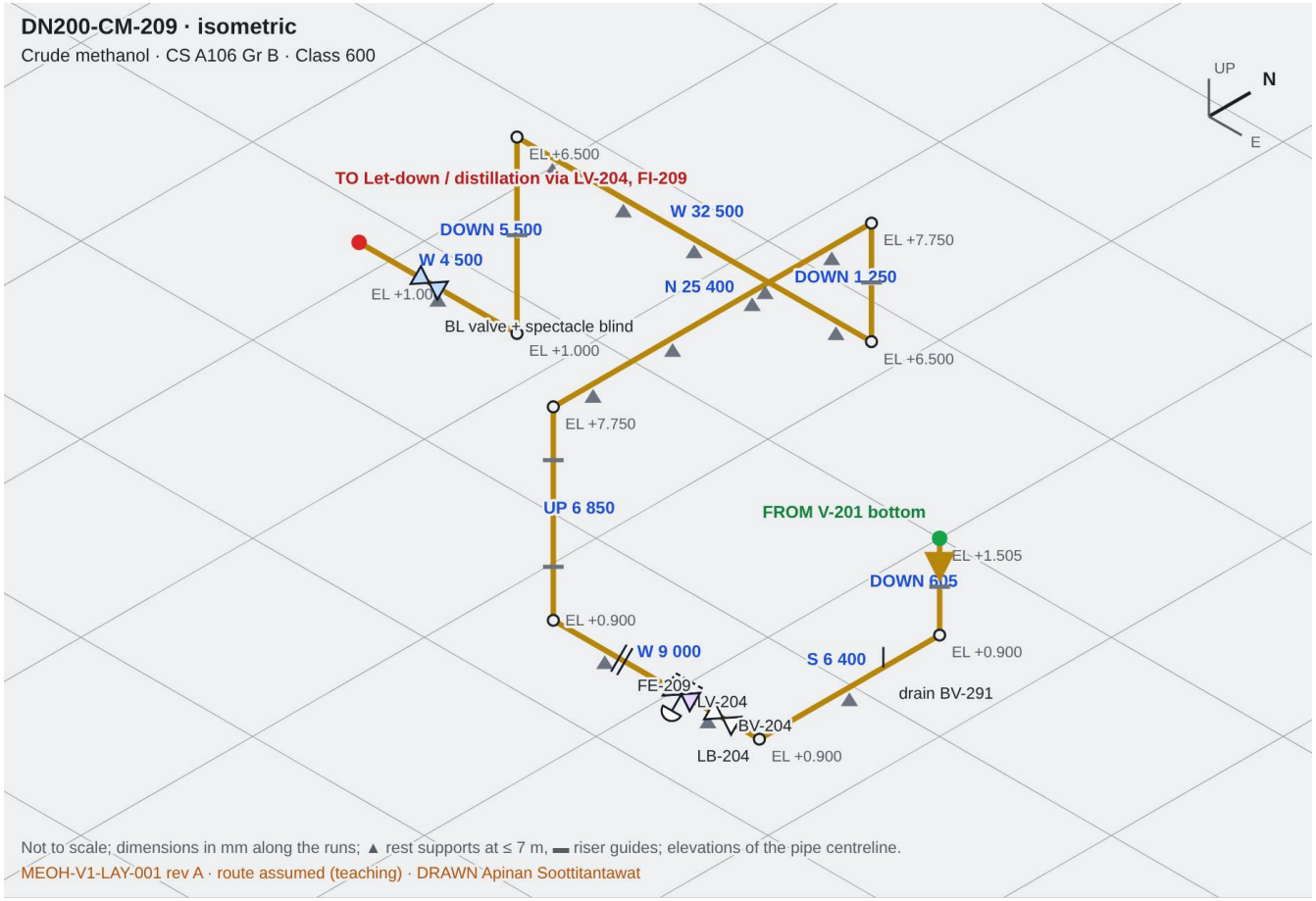
DN750-PG-202 — Converter feed (C-201 discharge): C-201 discharge → E-202 cold side inlet



CS A106 Gr B · Class 900 · 80.6 bar g / 48.6 °C

Item	Unit	Qty	Basis
DN750 pipe, CS A106 Gr B, wall 33 mm (heavy wall (B31.3 min. + CA, rounded up))	m	74.1	route 74.1 m (+5 % cutting) → 7 × 12 m random lengths
90° elbow, long radius (1.5 D)	ea	8	direction changes counted from the route
Branch tee	ea	0	one per branch line
Weld-neck flange Class 900	ea	6	two per in-line item + the two end connections
Gasket + stud-bolt set	set	4	one per flanged joint
Branch outlet (weldolet / sockolet)	ea	2	instrument, sample, drain and bypass connections
Pipe supports (shoes / rests)	ea	8	horizontal 62.2 m at ≤ 10 m spans (typical for DN750, assumed)
Riser guides	ea	5	vertical 11.9 m, one per 6 m (assumed)
Butt welds (shop + field, estimate)	ea	30	2 per elbow, 3 per tee, 1 per random-length joint, 1 per flange and branch
Block valve (gate / ball)	ea	1	BV-262
Orifice flow element + flange pair	ea	1	FE-201

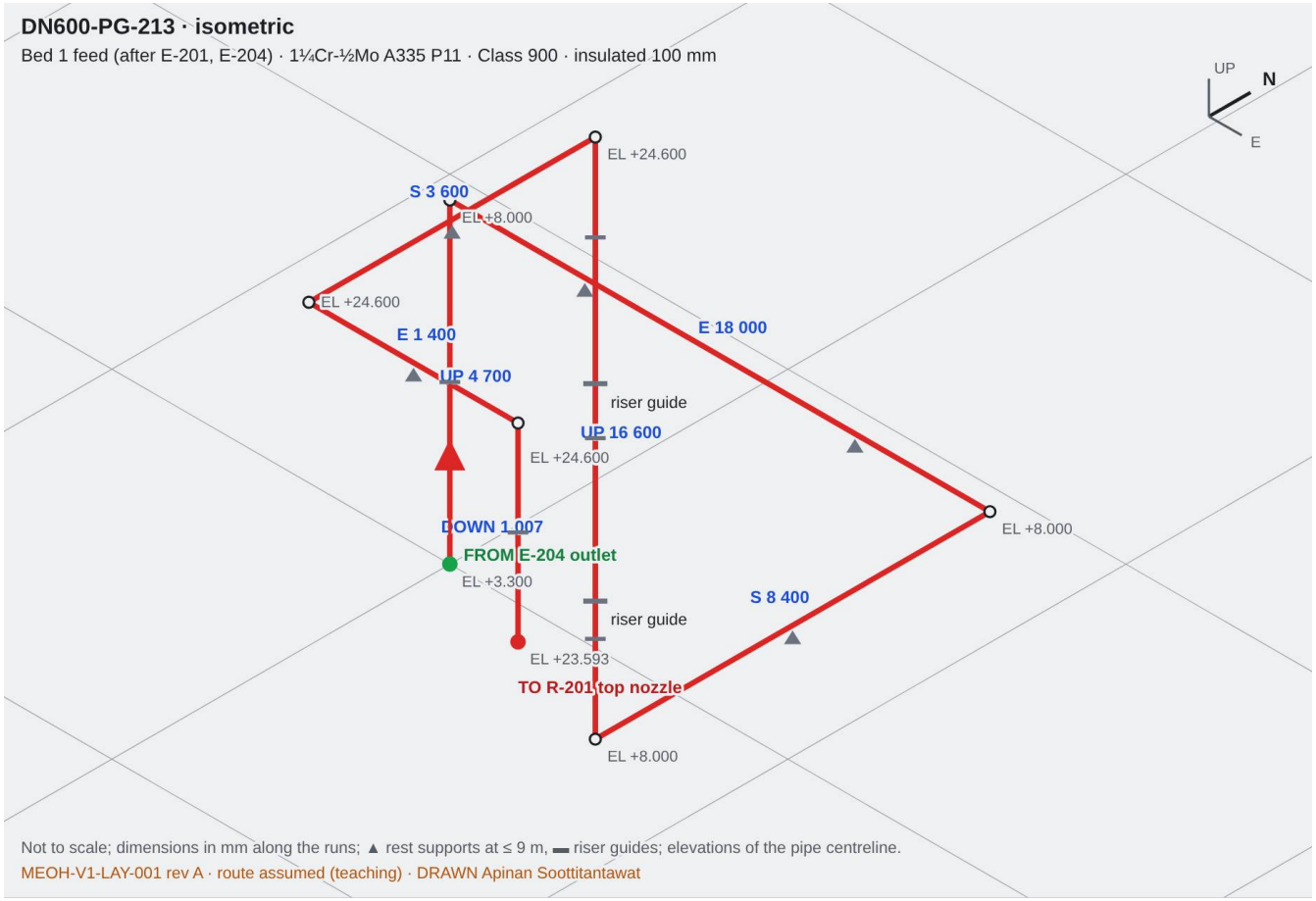
DN200-CM-209 — Crude methanol: V-201 bottom → Let-down / distillation via LV-204, FI-209



CS A106 Gr B · Class 600 · 75.0 bar g / 40.0 °C

Item	Unit	Qty	Basis
DN200 pipe, CS A106 Gr B, wall 12.7 mm (Sch 80)	m	92	route 92 m (+5 % cutting) → 9 × 12 m random lengths
90° elbow, long radius (1.5 D)	ea	8	direction changes counted from the route
Branch tee	ea	1	one per branch line
Weld-neck flange Class 600	ea	10	two per in-line item + the two end connections
Gasket + stud-bolt set	set	6	one per flanged joint
Branch outlet (weldolet / sockolet)	ea	1	instrument, sample, drain and bypass connections
Pipe supports (shoes / rests)	ea	13	horizontal 77.8 m at ≤ 7 m spans (typical for DN200, assumed)
Riser guides	ea	5	vertical 14.2 m, one per 6 m (assumed)
Butt welds (shop + field, estimate)	ea	38	2 per elbow, 3 per tee, 1 per random-length joint, 1 per flange and branch
Block valve (gate / ball)	ea	1	BV-204
Control valve	ea	1	LV-204
Bypass globe valve (parallel)	ea	1	LB-204
Orifice flow element + flange pair	ea	1	FE-209
Battery-limit valve + spectacle blind	ea	1	BL valve + spectacle blind

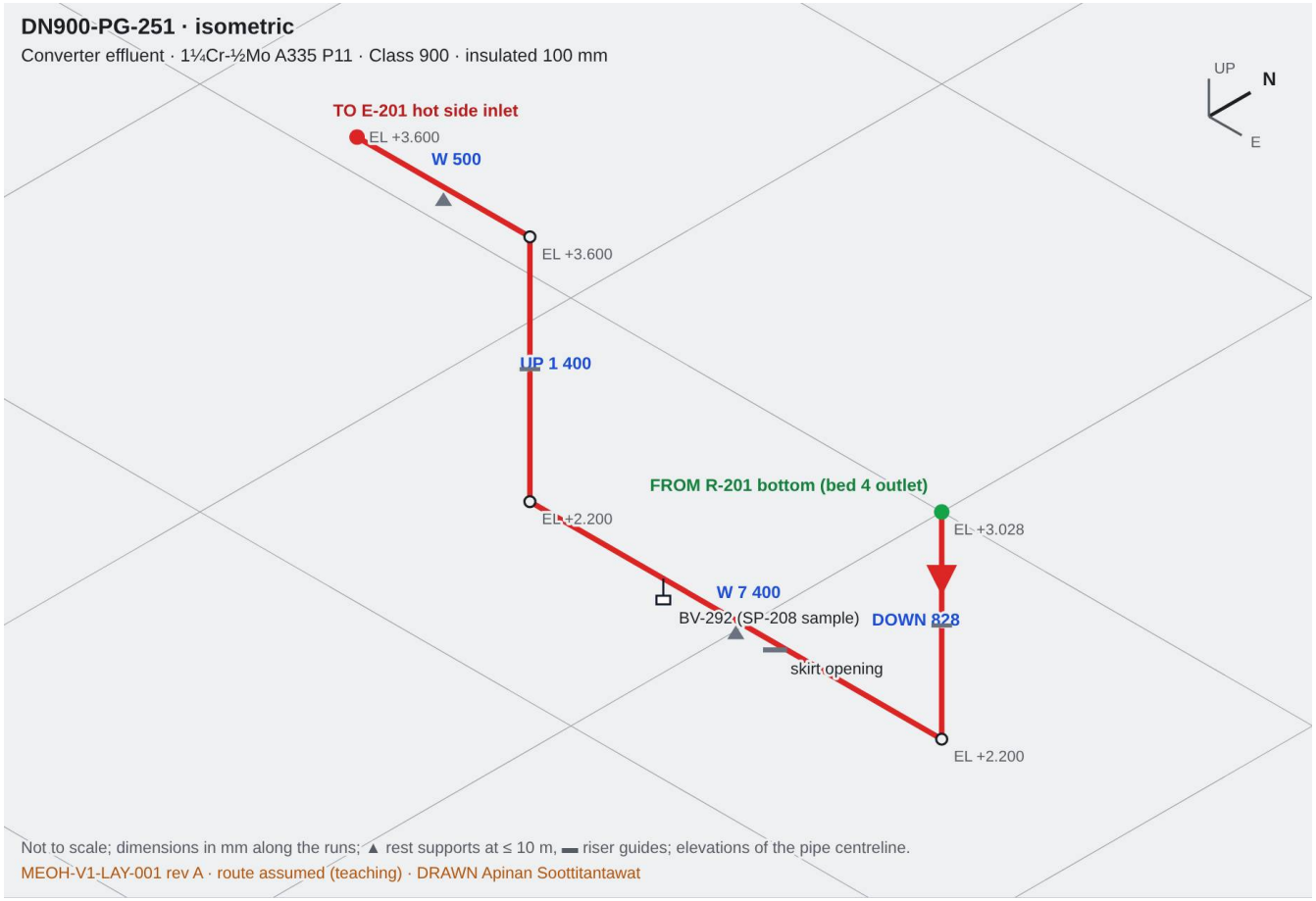
DN600-PG-213 — Bed 1 feed (after E-201, E-204): E-204 outlet → R-201 top nozzle



1¼Cr-½Mo A335 P11 · Class 900 · 79.7 bar g / 220.0 °C

Item	Unit	Qty	Basis
DN600 pipe, 1¼Cr-½Mo A335 P11, wall 30.96 mm (Sch 80)	m	53.7	route 53.7 m (+5 % cutting) → 5 × 12 m random lengths
90° elbow, long radius (1.5 D)	ea	6	direction changes counted from the route
Branch tee	ea	0	one per branch line
Weld-neck flange Class 900	ea	2	two per in-line item + the two end connections
Gasket + stud-bolt set	set	2	one per flanged joint
Branch outlet (weldolet / sockolet)	ea	0	instrument, sample, drain and bypass connections
Pipe supports (shoes / rests)	ea	5	horizontal 31.4 m at ≤ 9 m spans (typical for DN600, assumed)
Riser guides	ea	5	vertical 22.3 m, one per 6 m (assumed)
Butt welds (shop + field, estimate)	ea	18	2 per elbow, 3 per tee, 1 per random-length joint, 1 per flange and branch
Insulation 100 mm (hot service)	m	53.7	full route

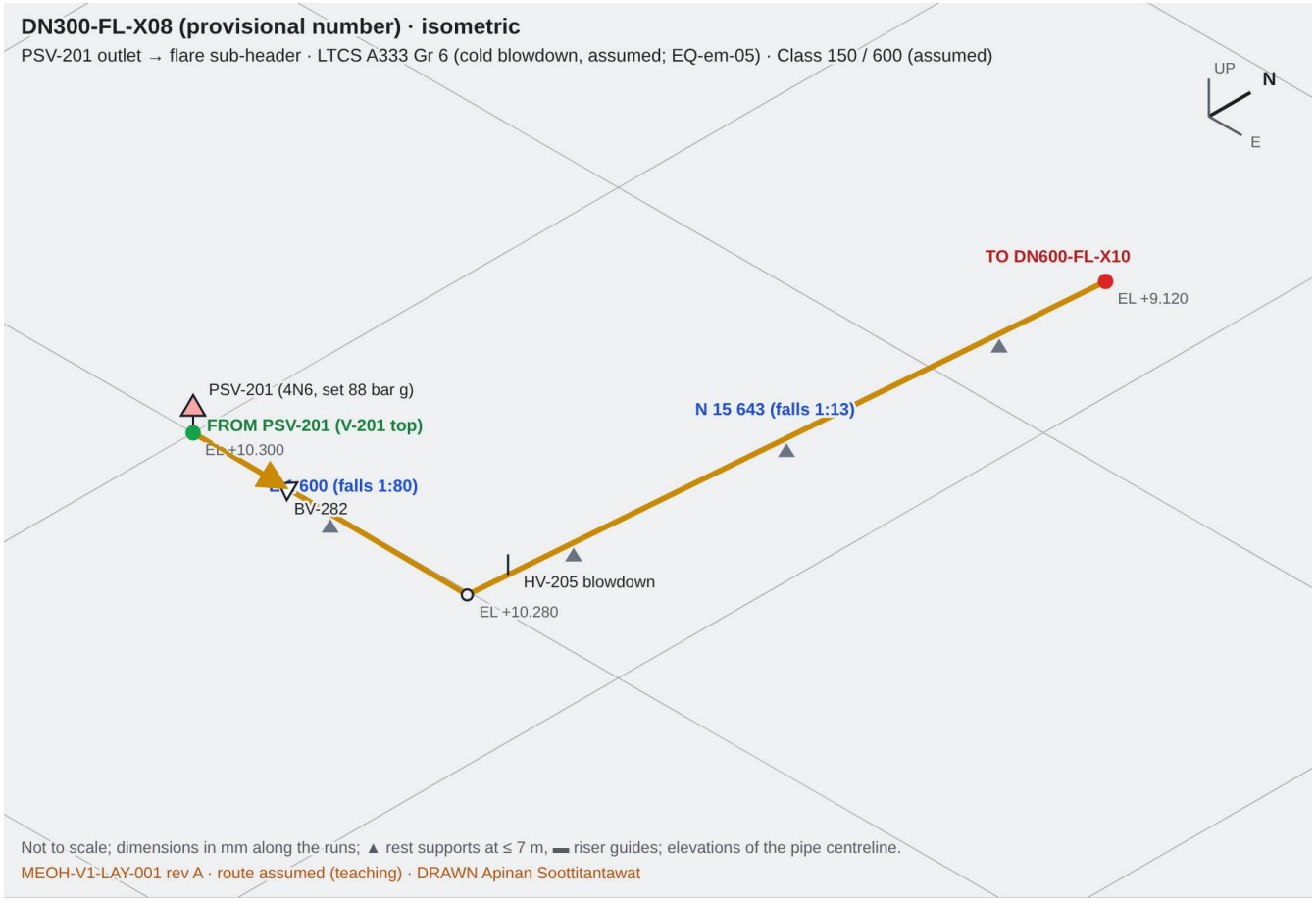
DN900-PG-251 — Converter effluent: R-201 bottom (bed 4 outlet) → E-201 hot side inlet



1¼Cr-½Mo A335 P11 · Class 900 · 77.4 bar g / 267.6 °C

Item	Unit	Qty	Basis
DN900 pipe, 1¼Cr-½Mo A335 P11, wall 43 mm (heavy wall (B31.3 min. + CA, rounded up))	m	10.1	route 10.1 m (+5 % cutting) → 1 × 12 m random lengths
90° elbow, long radius (1.5 D)	ea	3	direction changes counted from the route
Branch tee	ea	0	one per branch line
Weld-neck flange Class 900	ea	2	two per in-line item + the two end connections
Gasket + stud-bolt set	set	2	one per flanged joint
Branch outlet (weldolet / sockolet)	ea	1	instrument, sample, drain and bypass connections
Pipe supports (shoes / rests)	ea	2	horizontal 7.9 m at ≤ 10 m spans (typical for DN900, assumed)
Riser guides	ea	2	vertical 2.2 m, one per 6 m (assumed)
Butt welds (shop + field, estimate)	ea	9	2 per elbow, 3 per tee, 1 per random-length joint, 1 per flange and branch
Insulation 100 mm (hot service)	m	10.1	full route
Sample point	ea	1	BV-292 (SP-208 sample)

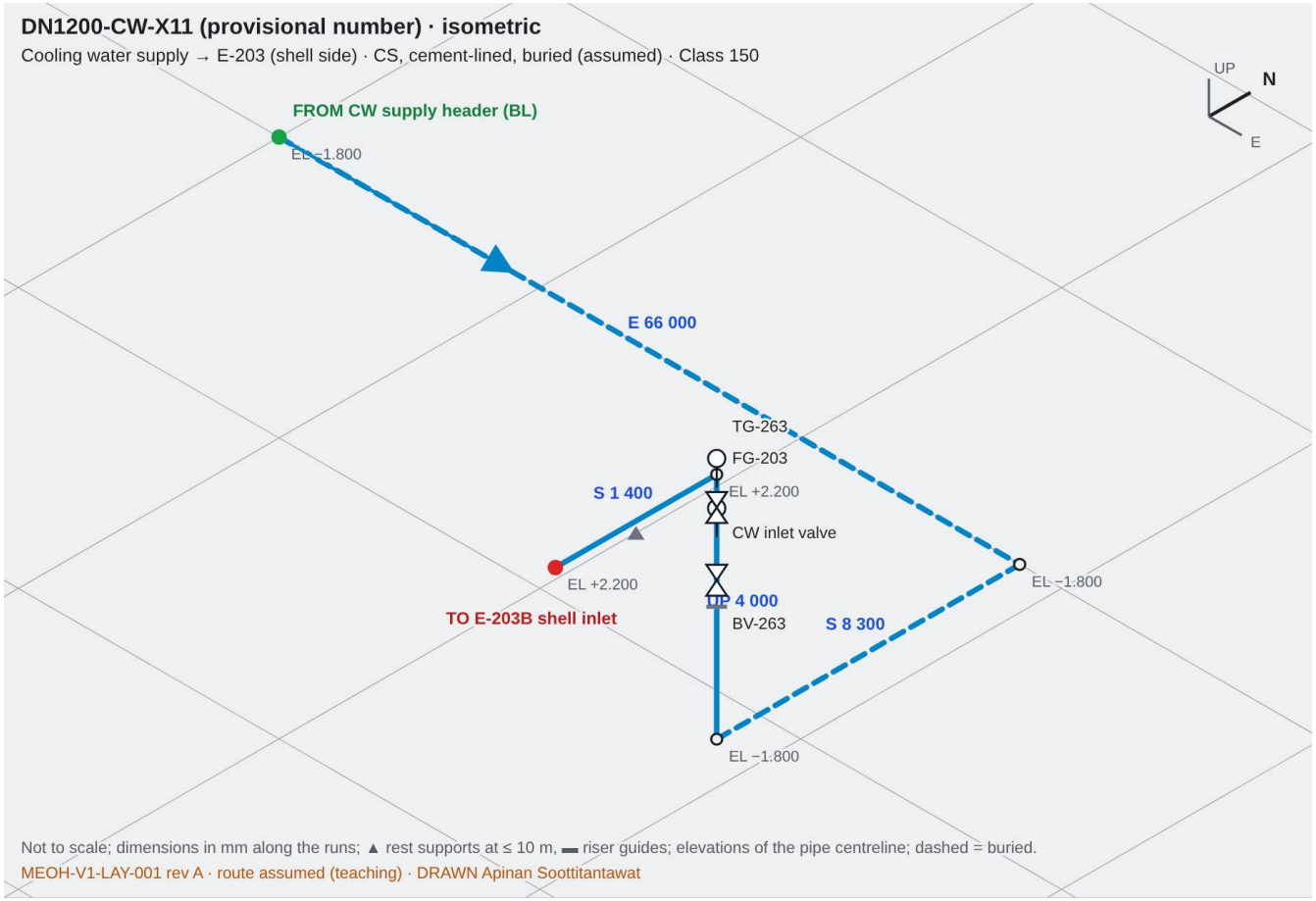
DN300-FL-X08 (provisional number) — PSV-201 outlet → flare sub-header: PSV-201 (V-201 top) → DN600-FL-X10



LTCS A333 Gr 6 (cold blowdown, assumed; EQ-em-05) · Class 150 / 600 (assumed) · 1.0 bar g / 40.0 °C

Item	Unit	Qty	Basis
DN300 pipe, LTCS A333 Gr 6 (cold blowdown, assumed; EQ-em-05) (Sch 40 (assumed))	m	17.2	route 17.2 m (+5 % cutting) → 2 × 12 m random lengths
90° elbow, long radius (1.5 D)	ea	1	direction changes counted from the route
Branch tee	ea	1	one per branch line
Weld-neck flange Class 150 / 600 (assumed)	ea	4	two per in-line item + the two end connections
Gasket + stud-bolt set	set	3	one per flanged joint
Branch outlet (weldolet / sockolet)	ea	0	instrument, sample, drain and bypass connections
Pipe supports (shoes / rests)	ea	4	horizontal 17.2 m at ≤ 7 m spans (typical for DN300, assumed)
Riser guides	ea	0	vertical 0 m, one per 6 m (assumed)
Butt welds (shop + field, estimate)	ea	10	2 per elbow, 3 per tee, 1 per random-length joint, 1 per flange and branch
Relief valve	ea	1	PSV-201 (4N6, set 88 bar g)
Block valve (gate / ball)	ea	1	BV-282

DN1200-CW-X11 (provisional number) — Cooling water supply → E-203 (shell side): CW supply header (BL) → E-203B shell inlet



CS, cement-lined, buried (assumed) · Class 150 · 4.0 bar g / 32.0 °C

Item	Unit	Qty	Basis
DN1200 pipe, CS, cement-lined, buried (assumed), wall 12.7 mm (STD (assumed))	m	79.7	route 79.7 m (+5 % cutting) → 7 × 12 m random lengths
90° elbow, long radius (1.5 D)	ea	3	direction changes counted from the route
Branch tee	ea	0	one per branch line
Weld-neck flange Class 150	ea	6	two per in-line item + the two end connections
Gasket + stud-bolt set	set	4	one per flanged joint
Branch outlet (weldolet / sockolet)	ea	2	instrument, sample, drain and bypass connections
Pipe supports (shoes / rests)	ea	0	buried line: bedding instead of supports
Riser guides	ea	1	vertical 4 m, one per 6 m (assumed)
Butt welds (shop + field, estimate)	ea	20	2 per elbow, 3 per tee, 1 per random-length joint, 1 per flange and branch
Block valve (gate / ball)	ea	2	BV-263, CW inlet valve

6 Layout checks

36 ✓, 0 to check, 5 ✗ of 41 checks, computed from layout.json, the EPC data and the model at the design point. Every ✗ carries an engineering query (section 7).

Check	Against	Value		Criterion (typical, assumed)	Query
Every equipment item of the EQL on the plot plan	MEOH-V1-ME-EQL-001	7 of 7 (R-201, E-201, E-202, E-203, E-204, V-201, C-201); E-203 as two shells A/B	✓	all 7 items	
Vessel and exchanger sizes as the datasheets	MEOH-V1-ME-DS-001…006	R-201 ID 4.81 m × T/T 16.62 m; V-201 ID 4.2 × 5.5 m; shells from the DS	✓	same as the EQL / DS	
Every P&ID line number routed in the layout, at the LSL size	MEOH-PID-001/002 · MEOH-V1-PE-LSL-001	11 of 11 numbered lines; size differences: none	✓	11 of 11	
Line numbers on the isometrics exist on the P&ID	MEOH-PID-001/002	4 of 6 isometrics (provisional: DN300-FL-X08, DN1200-CW-X11); 17 of 28 layout lines carry a provisional number	✗	every isometric line numbered on the P&ID	EQ-ly-01
Every field instrument of the INL has a tag box in the layout	MEOH-V1-IN-INL-001	56 of 56 (10 rows are DCS/SIS functions without a field device)	✓	all field devices	
Every hand valve of tabs/v3d/hv.json placed (on its P&ID line where it has one)	tabs/v3d/hv.json	16 of 16 placed; 8 of 8 on their line	✓	all	
Percent field valves and field switches of the Field tab placed	Field tab (FIELDV, SWITCHES)	4 of 4 field valves, 3 of 3 switches	✓	all	
Local gauges read the model at the design point, inside 25–75 % of their range	model snapDesign · gauge ranges	19 of 19 readings finite; 7 of 7 dials inside 25–75 %	✓	normal reading 25–75 % of the dial range (typical gauge selection, assumed)	
No equipment footprints overlap	plot plan	none	✓	clear footprints	
Clear distance between process equipment	plot plan	closest E-203A / E-203B 1.6 m (exchangers side by side, ≥ 1.5 m); 0 pairs short	✓	≥ 3 m (1.5 m between exchanger shells): clear distance between process equipment for access and maintenance (typical, assumed)	
C-201 (process-gas compressor) to other process equipment	plot plan	closest V-201 / C-201-M 22.5 m	✓	≥ 9 m: process-gas compressor to other process equipment / reactors (typical of insurer spacing tables, e.g. GAP.2.5.2 / CCPS siting guidelines; assumed)	
Control room CCR (off-plot) to the nearest process equipment	plot plan · key plan	80.4 m to C-201	✓	≥ 60 m: non-blast-resistant occupied building from process equipment (typical spacing table; a building inside this needs an API RP 752 blast study; assumed)	
Substation SS-201 to process equipment, outside every hazardous zone	plot plan · MEOH-V1-EE-HAC-001	41.2 m to C-201; zones overlapping: none	✓	≥ 30 m: substation from process equipment, outside every hazardous zone (typical, assumed)	
Ignition sources (CCR, SS-201 with the VSD) from H_2 equipment	plot plan	CCR 80.4 m (C-201); SS-201 41.2 m (C-201)	✓	≥ 30 m: ignition sources (substation, control room, VSD) from high-pressure H_2 equipment (typical, assumed)	
CCR and SS-201 upwind of the unit (prevailing wind from the south-west)	meta.wind (assumed)	bearing from the unit: CCR 229°, SS-201 258° (upwind 225°)	✓	within ±45°: control room and substation inside ±45° of the upwind direction (assumed)	
Flare stack (off-plot) to process equipment	key plan	169 m to R-201	✓	≥ 90 m: flare stack from process equipment (radiation, API 521 typical; off-plot; assumed)	

Check	Against	Value		Criterion (typical, assumed)	Query
Access roads ≥ 6 m wide; equipment clear of the road edge	plot plan	4 roads, narrowest 6 m; closest equipment E-203A 13 m from RD-S	✓	access roads ≥ 6 m wide; equipment ≥ 3 m from the road edge (typical, assumed)	
Pipe-rack width for the lines it carries (+25 % spare), lanes inside the rack	PR-201 · lane list	lower tier 4.99 m needed of 8 m; upper tier 2.63 m needed of 8 m	✓	rack width: Σ(OD + 2 × insulation + 150 mm) per tier + 25 % future space (typical, assumed)	
Pipe-rack edge to equipment	plot plan	7.2 m to E-202	✓	≥ 3 m: pipe-rack edge to equipment (access, fire; typical, assumed)	
Tube-bundle withdrawal space at the channel end	EQL tube lengths · plot plan	E-202 99/7.6 m; E-203A 99/13.7 m; E-203B 99/13.7 m; E-204 99/7.6 m	✓	tube-bundle withdrawal space: tube length + 1.5 m clear at the channel end (typical, assumed)	
C-201 casing and motor under the shelter crane; laydown ≥ motor + 1 m each side	SH-201 · LA-201	casing and motor inside the crane area; laydown 7.6 × 9.5 m for a 5.4 × 3.2 m motor	✓	crane lift and laydown without a mobile crane (assumed)	
Crane hardstand for R-201 catalyst handling	CH-201	8.2 m from the converter centre	✓	≤ 15 m: crane hardstand within 15 m of the converter centre for catalyst bins and top-head lifts (assumed)	
Stair pitch	stairs	30.3–41.9° over 9 flights	✓	30–45° (industrial stair pitch 30–45° (typical, e.g. ISO 14122-3 / OSHA 1910.25))	
Ladder flights	ladders	longest 4.5 m (9 flights)	✓	≤ 6 m: fixed ladder flight ≤ 6 m between rest landings (ISO 14122-4 typical; assumed)	
Headroom under platforms	platforms	least 2.2 m (under PF-C1)	✓	≥ 2.1 m: clear headroom under platforms and pipes over walkways (typical, assumed)	
Escape from every platform: two ways down, or a dead end ≤ 7.5 m	platforms · stairs · ladders	7 of 8 platform groups pass; failing: PF-R2 + BR-E1 + PF-E1 (2 ways down, dead end 9.4 m at PF-E1 (only route through BR-E1))	✗	dead end ≤ 7.5 m, travel ≤ 25 m (dead-end travel on a platform before two ways down are available ≤ 7.5 m (typical company standard, assumed))	EQ-ly-03
From the foot of every stair and ladder to an access road	grade	longest 32.1 m (ST-R1)	✓	≤ 45 m: travel at grade from the foot of a stair or ladder to an access road ≤ 45 m (typical, assumed)	
Hand valves, field valves and switches within reach of grade or a platform	operables	23 ✓ · 0 to check · 0 ✗ of 23	✓	hand valves operable 0.3–1.8 m above the standing surface; to 2.1 m with care; above needs a platform, extension spindle or chain wheel (typical ergonomics, assumed)	
Local gauges, stem indicators, LG-204 and the C-201 panel readable	instruments	19 ✓ · 0 to check · 0 ✗ of 19	✓	local dial gauges and stem indicators read 0.5–2.0 m above the standing surface; to 3.0 m from a distance (assumed); magnetic flag level gauge legible 0.3–4.0 m above the standing surface (assumed)	
LG-204 in sight from the LV-204 bypass LB-204 (manual level control)	LG-204 · LB-204	5 m, glass facing S (towards the operator)	✓	≤ 10 m and facing: the operator at LB-204 sees LG-204 within 10 m, facing the glass (manual level control on the bypass; assumed)	
Emergency blowdown HV-205 reached from grade or a stair-served platform	HV-205 (manual, MEOH-PID-002)	on PF-V1 EL +9.2 (ladder LD-V2)	✗	an emergency action must not need a ladder climb onto the inventory vessel (assumed)	EQ-ly-04
Flare and blowdown lines free-draining (no pockets) towards the flare header	DN300-FL-X08, X09, X10	3 lines fall continuously; PSV-201 outlet EL +10.3 above the header EL +9.12	✓	API 521 practice: relief and flare lines self-draining, no liquid pockets	

Check	Against	Value		Criterion (typical, assumed)	Query
C-201 suction line DN900-PG-205 drains back to V-201 (no low point at the C-201 nozzle)	DN900-PG-205 profile	low point EL +3.4 at the C-201 suction nozzle; low-point drain fitted; V-201 outlet EL +9.99	✗	compressor suction lines slope back to the knock-out drum; no liquid can collect at the machine (typical)	EQ-ly-02
Upward two-phase legs E-203 → V-201 at a mist / annular gas velocity	DN900-PG-X07, MEOH-V1-ME-DS-002 vapour flow	13.3 m/s at design (7.29 m³/s in DN900 ID 836 mm); line falls 0.1 m to the V-201 inlet	✓	≥ 5 m/s: upward two-phase (gas-dominated) flow acceptable when the gas velocity ≥ 5 m/s (mist / annular flow; typical, assumed)	
V-201 elevation = relief basis fire case	MEOH-V1-PE-RLB-001 (fire, S4)	bottom tangent EL +3; RLB 3 m	✓	same elevation (wetted area)	
AIT-208 heated sample line length	MEOH-V1-EE-LL-001 L11	27.2 m (SP-208 → AH-201, rectilinear)	✓	≤ 30 m: AIT-208 heated sample line ≤ 30 m (heat-tracing basis of MEOH-V1-EE-LL-001 L11)	
Substation → C-201 motor cable route	MEOH-V1-EE-LL-001 (cable example)	85 m (rectilinear × 1.3)	✓	≤ 150 m: substation → C-201 cable route ≤ 150 m (MEOH-V1-EE-LL-001 cable example; route = rectilinear × 1.3, assumed)	
Local gauges and hand valves of the layout on the P&ID	MEOH-PID-002 rev A · EQ-ec-05	8 local gauges and 16 hand valves are not drawn on rev A	✗	every operator-read gauge and every hand valve on the P&ID	EQ-ly-05
R-201 bed nozzles (quench N3A–C, bed-inlet TE, multipoint T1–T4) above their bed platform	DS-001 nozzles · platforms	0.72–1.61 m above the floor	✓	0.5–2.0 m (work height, assumed)	
R-201 catalyst dump nozzles clear of grade	DS-001 D1/D2 · skirt	bottom head EL +3.63, dump flange ≈ EL +3.33	✓	≥ 2.0 m for the dump chute and bins (assumed)	
R-201 top head and manways above the top platform	DS-001 M1/M2 · PF-R5	apex 0.59 m above the floor	✓	0.3–1.2 m (assumed)	

Escape from the platforms

Platform group	EL	Ways down	Travel (m)	Dead end (m)	
PF-C1	2.40	stair ST-C1, ladder LD-C1	5.8	0.0	✓
PF-V2	2.60	stair ST-V1	7.4	7.4	✓
PF-R4	9.00	stair ST-R3, ladder LD-R2	10.4	0.0	✓
PF-V1	9.20	ladder LD-V2	7.3	7.3	✓
PF-R3	13.30	stair ST-R4, ladder LD-R3	10.4	0.0	✓
PF-R2 + BR-E1 + PF-E1	16.80	stair ST-R5, ladder LD-R4	17.6	9.4	✗
PF-R1	19.60	stair ST-R6, ladder LD-R5	10.4	0.0	✓
PF-R5	22.40	stair ST-R7, ladder LD-R6	10.4	0.0	✓

Access to hand valves, switches and local gauges

Tag	Item	Kind	EL	Stands on	Height (m)	
BV-202	FV-202 upstream block valve on the make-up syngas line	hand valve	1.40	grade	1.40	✓
BV-203	PV-203 upstream block valve on the purge line	hand valve	1.60	grade	1.60	✓
BV-204	LV-204 upstream block valve on the crude methanol let-down	hand valve	1.30	grade	1.30	✓
BV-221	TV-221 upstream block valve, bed 2 cold shot	hand valve	1.60	grade	1.60	✓
BV-241	TV-241 upstream block valve, bed 4 cold shot	hand valve	1.60	grade	1.60	✓
BV-261	C-201 suction block valve	hand valve	4.20	PF-C1	1.80	✓
BV-262	C-201 discharge block valve	hand valve	4.20	PF-C1	1.80	✓
BV-263	E-203 cooling-water supply block valve	hand valve	0.90	grade	0.90	✓
BV-264	E-203 cooling-water return block valve	hand valve	1.20	grade	1.20	✓
BV-271	LG-204 gauge glass upper (vapour) isolation valve	hand valve	4.30	PF-V2	1.70	✓
BV-272	LG-204 gauge glass lower (liquid) isolation valve	hand valve	3.15	PF-V2	0.55	✓
BV-281	PSV-201 inlet isolation valve, car-sealed open	hand valve	9.85	PF-V1	0.65	✓
BV-282	PSV-201 outlet isolation valve to flare, car-sealed open	hand valve	10.30	PF-V1	1.10	✓
BV-283	HV-205 blowdown upstream block valve, locked open	hand valve	10.80	PF-V1	1.60	✓
BV-291	V-201 bottom drain valve, locked closed	hand valve	1.20	grade	1.20	✓
BV-292	Converter outlet sample valve, locked closed between samples	hand valve	1.30	grade	1.30	✓
HV-205	HV-205 loop blowdown to flare (manual, emergency depressuring)	field valve (%)	10.30	PF-V1	1.10	✓
LB-204	LV-204 hand bypass (globe, normally shut)	field valve (%)	1.30	grade	1.30	✓
PB-203	PV-203 purge hand bypass (globe, normally shut)	field valve (%)	1.60	grade	1.60	✓
BV-231	TV-231 upstream block valve (normally 100 % open)	field valve (%)	1.60	grade	1.60	✓
C-201 LCS	C-201 local control station (RUN / STOP)	switch	1.30	grade	1.30	✓
Steam block valve	E-204 start-up steam block valve	switch	1.70	grade	1.70	✓
CW inlet valve	E-203 cooling water inlet valve	switch	1.70	grade	1.70	✓
PG-202	Make-up syngas pressure at the BL (upstream of FV-202)	local gauge	1.50	grade	1.50	✓
PG-261	C-201 suction pressure	local gauge	3.90	PF-C1	1.50	✓
PG-262	C-201 discharge pressure	local gauge	3.90	PF-C1	1.50	✓
PG-203	V-201 pressure (top)	local gauge	10.00	PF-V1	0.80	✓
LG-204	V-201 level gauge (magnetic flag type, 0–100 % = EL +3.30–5.47 m)	level gauge	4.38	PF-V2	1.78	✓
TG-263	E-203 cooling water supply temperature	local gauge	2.00	grade	2.00	✓
TG-264	E-203 cooling water return temperature	local gauge	1.60	grade	1.60	✓
FG-203	E-203 cooling water flow indicator (% of normal)	local gauge	1.25	grade	1.25	✓
LP-201	C-201 local panel: speed, motor power, vibration; local STOP	local display	1.60	grade	1.60	✓
FV-202	FV-202 local stem position indicator	stem indicator	1.60	grade	1.60	✓

Tag	Item	Kind	EL	Stands on	Height (m)	
XV-202	XV-202 local stem position indicator	stem indicator	1.60	grade	1.60	✓
PV-203	PV-203 local stem position indicator	stem indicator	1.70	grade	1.70	✓
LV-204	LV-204 local stem position indicator	stem indicator	1.50	grade	1.50	✓
TV-211	TV-211 local stem position indicator	stem indicator	1.80	grade	1.80	✓
TV-221	TV-221 local stem position indicator	stem indicator	1.80	grade	1.80	✓
TV-231	TV-231 local stem position indicator	stem indicator	1.80	grade	1.80	✓
TV-241	TV-241 local stem position indicator	stem indicator	1.80	grade	1.80	✓
HV-214	HV-214 local stem position indicator	stem indicator	1.70	grade	1.70	✓
HV-205	HV-205 local stem position indicator	stem indicator	10.60	PF-V1	1.40	✓

Pipe-rack width (worked example)

Tier	Lines: OD + 2 × insulation + 150 mm (m)	Σ (m)	× 1.25 (m)	Width (m)	
lower EL +6.5	DN900-PG-205 1.06, DN750-PG-202 0.91, DN1000-PG-X05 1.33, DN200-CM-209 0.37, DN150-PU-203 0.32	3.99	4.99	8.00	✓
upper EL +9	DN600-FL-X10 0.76, DN200-HS-X13 0.53, DN100-SC-X14 0.36, DN80-NG-X15 0.24, DN50-IA-X16 0.21	2.10	2.63	8.00	✓

7 Engineering queries

No.	From	To	Issue	Proposed
EQ-ly-01	MEOH-V1-LAY-001	MEOH-PID-001 / MEOH-PID-002 · MEOH-V1-PE-LSL-001	The layout routes 28 lines but only 11 carry a P&ID line number. Two of the six isometrics (DN300-FL-X08, DN1200-CW-X11) and 15 other lines use provisional numbers: the 8 unnumbered process lines of EQ-ep-06 (bed-1 branch, TV-211 bypass, E-201 → E-204, E-201 → E-202, E-202 → E-203, E-203A → B, E-203 → V-201, the PSV-201 and HV-205 flare lines) and every utility line (flare sub-header, CW supply and return, HP steam and condensate to E-204, N_2, instrument air, V-201 drain), which the P&IDs show only as arrows.	Number every line on MEOH-PID-001/002 rev B with the next free numbers of each service (PG-2xx, FL-2xx, CW-2xx, HS-2xx …), add the utility lines with their sizes to the line sizing list, and reissue the isometrics with the final numbers.
EQ-ly-02	MEOH-V1-LAY-001	MEOH-PID-002 / C-201 vendor (DS-007)	DN900-PG-205 rises from the V-201 top (EL +9.99), runs on the rack at EL +6.5 and drops 3.1 m to the top suction nozzle of the grade-mounted C-201 (EL +3.40): the low point of the suction line is the machine. The loop gas leaves V-201 saturated at 40 °C, so condensate (cooler ambient, a demister upset) runs to C-201; V-201 is the only knock-out drum and LSHH-204 protects only the V-201 level. The layout fits a low-point drain only.	Add a suction drain pot (or a small knock-out drum) at the C-201 inlet with a level gauge, LSH alarm and drain to the closed drain; insulate PG-205; or mount C-201 on a table-top so the suction line falls back to V-201 (1:500). Confirm the vendor's liquid tolerance.
EQ-ly-03	MEOH-V1-LAY-001	Structural / platform layout (E-201)	The E-201 top platform PF-E1 (EL +16.8) is reached only over the bridge BR-E1 from the R-201 bed 2 platform: from its far corner an operator walks 9.4 m before two ways down are available, above the 7.5 m dead-end limit used here (assumed). A fire at the bridge or the quench risers would trap him.	Add a caged ladder from PF-E1 down the west side of E-201 with a rest landing on the STR-E201 structure (EL +10.0), or extend the structure as a second stair; recheck when the E-201 structure is detailed.
EQ-ly-04	MEOH-V1-LAY-001	MEOH-PID-002 note 6 · EQ-er-06 · HAZOP R15	HV-205 is a manual blowdown valve. Its line must fall continuously to the flare sub-header (EL +9), so the valve sits high on the V-201 top platform PF-V1 (EL +9.2), reached only by two ladder flights. In the emergency HV-205 is meant for (fire, leak) the operator would have to climb onto the top of the loop inventory vessel.	Make HV-205 a remotely operated, fail-open blowdown valve (BDV) with a hand switch in the CCR and at a safe location at grade (as EQ-er-06 / HAZOP R15 propose); keep a local hand-wheel only for maintenance. If it stays manual, move it to a stair-served platform on the rack side with the line still free-draining.
EQ-ly-05	MEOH-V1-LAY-001	MEOH-PID-002 rev A · EQ-ec-05	The layout places 8 local gauges an outside operator reads (PG-202, PG-261, PG-262, PG-203, LG-204, TG-263, TG-264, FG-203), 16 hand valves (BV-202 … BV-292, from the field model) and the battery-limit valves; none is drawn on MEOH-PID-001/002 rev A, and their ranges, sizes and car-seal status exist only in the layout and the simulator.	Show every local gauge (with range), every block, bypass, drain and isolation valve (with NO / CSO / LC) and the BL valves with spectacle blinds on rev B; add them to the piping bulk and the valve list (see EQ-ec-05).

Criteria used (all typical practice, assumed)

Rule	Value	Source
eq	v 3	clear distance between process equipment for access and maintenance (typical, assumed)
hx	v 1.5	clear distance between exchanger shells side by side (typical, assumed)
comp	v 9	process-gas compressor to other process equipment / reactors (typical of insurer spacing tables, e.g. GAP.2.5.2 / CCPS siting guidelines; assumed)
ign	v 30	ignition sources (substation, control room, VSD) from high-pressure H_2 equipment (typical, assumed)
ccr	v 60	non-blast-resistant occupied building from process equipment (typical spacing table; a building inside this needs an API RP 752 blast study; assumed)
ss	v 30	substation from process equipment, outside every hazardous zone (typical, assumed)
flare	v 90	flare stack from process equipment (radiation, API 521 typical; off-plot; assumed)
rack	v 3	pipe-rack edge to equipment (access, fire; typical, assumed)
road	v 3, w 6	access roads ≥ 6 m wide; equipment ≥ 3 m from the road edge (typical, assumed)

Rule	Value	Source
reach	lo 0.3, hi 1.8, max 2.1	hand valves operable 0.3–1.8 m above the standing surface; to 2.1 m with care; above needs a platform, extension spindle or chain wheel (typical ergonomics, assumed)
read	lo 0.5, hi 2, max 3	local dial gauges and stem indicators read 0.5–2.0 m above the standing surface; to 3.0 m from a distance (assumed)
glass	lo 0.3, hi 4	magnetic flag level gauge legible 0.3–4.0 m above the standing surface (assumed)
stair	lo 30, hi 45	industrial stair pitch 30–45° (typical, e.g. ISO 14122-3 / OSHA 1910.25)
ladder	v 6	fixed ladder flight ≤ 6 m between rest landings (ISO 14122-4 typical; assumed)
headroom	v 2.1	clear headroom under platforms and pipes over walkways (typical, assumed)
pf_struct	0.2	
dead	v 7.5	dead-end travel on a platform before two ways down are available ≤ 7.5 m (typical company standard, assumed)
travel	v 25	travel on a platform to the nearest way down ≤ 25 m (typical, assumed)
gradeEsc	v 45	travel at grade from the foot of a stair or ladder to an access road ≤ 45 m (typical, assumed)
bundle	v 1.5	tube-bundle withdrawal space: tube length + 1.5 m clear at the channel end (typical, assumed)
rackSpare	v 0.25, gap 0.15	rack width: Σ(OD + 2 × insulation + 150 mm) per tier + 25 % future space (typical, assumed)
crane	v 15	crane hardstand within 15 m of the converter centre for catalyst bins and top-head lifts (assumed)
sample	v 30	AIT-208 heated sample line ≤ 30 m (heat-tracing basis of MEOH-V1-EE-LL-001 L11)
cable	v 150	substation → C-201 cable route ≤ 150 m (MEOH-V1-EE-LL-001 cable example; route = rectilinear × 1.3, assumed)
gauge	lo 25, hi 75	normal reading 25–75 % of the dial range (typical gauge selection, assumed)
twoPhase	v 5	upward two-phase (gas-dominated) flow acceptable when the gas velocity ≥ 5 m/s (mist / annular flow; typical, assumed)
lgSight	v 10	the operator at LB-204 sees LG-204 within 10 m, facing the glass (manual level control on the bypass; assumed)
wind	v 45	control room and substation inside ±45° of the upwind direction (assumed)